

HW#4

Deep Learning for Computer Vision (EN525.733.8VL)

Due date: March 5th, 2026

Q.1: Use a 3-layer perceptron autoencoder to perform dimensionality reduction (compression) on MNIST dataset.

Download the MNIST dataset (training, validation and test datasets are provided)

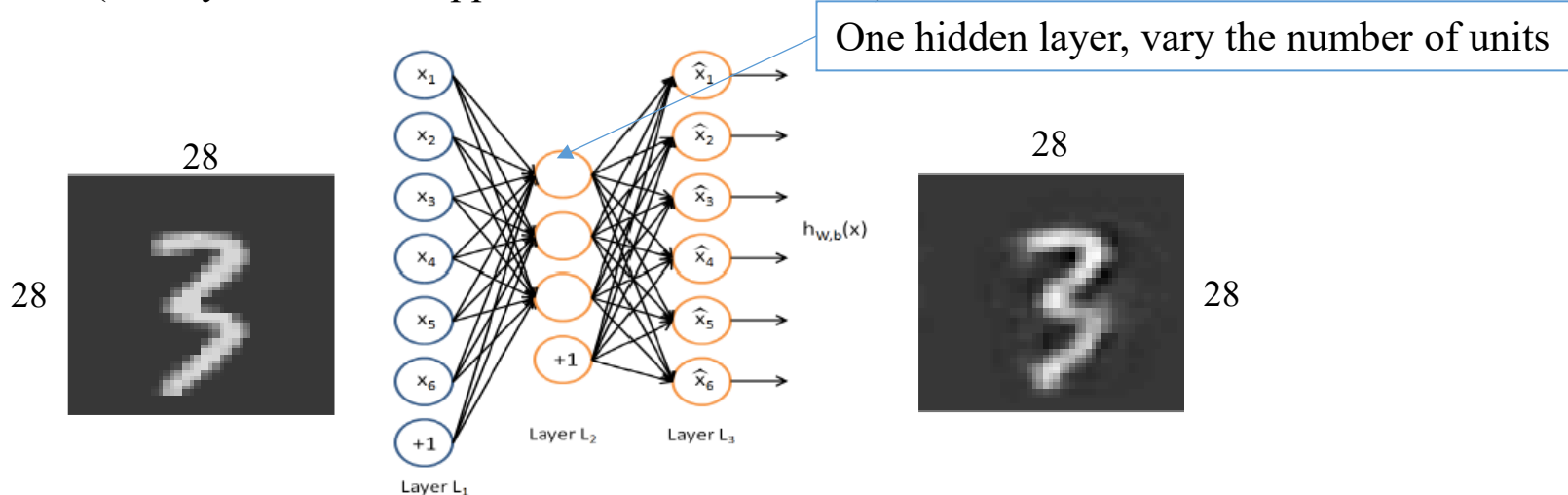
<http://yann.lecun.com/exdb/mnist/>

1. Read my Autoencoder notes + Also check these two links

<https://www.jeremyjordan.me/autoencoders/>

<https://medium.com/edureka/autoencoders-tutorial-cfdcebdefe37>

1. Train your MLPs with one hidden layer (see figure below) with different number of hidden units (try 5, 10, 30, 60, 120 etc.). Evaluate the reconstruction performance (Mean Squared Error (MSE)) for an integer 3 with different hidden unites (5, 10, 30, 60, 120). **Plot learning curve vs iteration and plot the curves for the validation and testing datasets (treat test set like validation dataset for plotting).**
2. For 10 and 20 hidden unites display the set of weights converging to each neuron as an image of size 28x28 (see my course notes pp.78+79 on autoencoder).



THE MNIST DATABASE

of handwritten digits

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[Corinna Cortes](#), Google Labs, New York

[Christopher J.C. Burges](#), Microsoft Research, Redmond

The MNIST database of handwritten digits, available from this page, has a training set of 60,000 examples, and a test set of 10,000 examples. It is a subset of a larger set available from NIST. The digits have been size-normalized and centered in a fixed-size image.

It is a good database for people who want to try learning techniques and pattern recognition methods on real-world data while spending minimal efforts on preprocessing and formatting.

Four files are available on this site:

train-images-idx3-ubyte.gz :	training set images (9912422 bytes)
train-labels-idx1-ubyte.gz :	training set labels (28881 bytes)
t10k-images-idx3-ubyte.gz :	test set images (1648877 bytes)
t10k-labels-idx1-ubyte.gz :	test set labels (4542 bytes)

The MNIST dataset with 60,000 training images and labels and 10,000 testing images and labels. For validation set use 20% of the training dataset and rest for training.